

PRIMITIVE

$$f(x) = \frac{1}{x^2 + 1}$$

$$f'(x) = \frac{1}{(x^2 + 1)^2} \cdot 2x$$

CERCO F T.C. $F'(x) = f(x)$

$$F(x) = \arctan x$$

DEF. SIA $f: A \rightarrow \mathbb{R}$.

UNA FUNZIONE $F: A \rightarrow \mathbb{R}$ SI CHIAMA
PRIMITIVA DI f SE:

- F DERIVABILE

$$- F'(x) = f(x) \quad \forall x \in A$$

es) $\arctan x$ È UNA PRIMITIVA DI $\frac{1}{x^2 + 1}$

es) $\arctan x + 3$ È UNA PRIMITIVA DI $\frac{1}{x^2 + 1}$

NOTAZIONE:

$$\int f(x) dx = \{ F: A \rightarrow \mathbb{R} : F' = f \}$$

↑
INTEGRALE INDEFINITO DI f

$$\log(x^2+1) \rightsquigarrow \frac{1}{x^2+1} \cdot 2x$$

$$\log\left(\frac{x^2+1}{2x}\right) \text{ NO}$$

$$\frac{1}{2x} \log(x^2+1) \rightarrow \frac{1}{x^2+1} \cdot \frac{2x}{2x} - \frac{1}{2x^2} \log(x^2-1)$$

$$(\arctan x + C)' = \frac{1}{x^2+1}$$

$$F'(x) = \frac{1}{x^2+1}$$

$$(F(x) = \arctan x)' = 0 \quad \forall x \in \mathbb{R}$$

$$\Rightarrow F(x) = \arctan x = C$$

$$\Rightarrow F(x) = \arctan x + C$$

OSSERVAZIONE. $f: I \rightarrow \mathbb{R}$, I INTERVALLO

F PRIMITIVA DI f , ALLORA

$$\int f(x) dx = \{F + C : C \in \mathbb{R}\}$$

ABUSO DI NOTAZIONE $\curvearrowright = F(x) + C$

es)

$$f: (-\infty, 0) \cup (0, +\infty) \rightarrow \mathbb{R}$$

$$f(x) = \frac{1}{x}$$

$$F(x) = \begin{cases} \log x & x > 0 \\ \log(-x) & x < 0 \end{cases}$$

$$(\log x)' = \frac{1}{x}, \quad -1 = \frac{1}{x}$$

$$F(x) = \log|x|$$

$$\Rightarrow \int \frac{1}{x} dx = \log|x| + C$$

• DUE QUESTIONI

1) $\exists$ PRIMITIVA?

2) DETERMINABILE

• TEOREMA

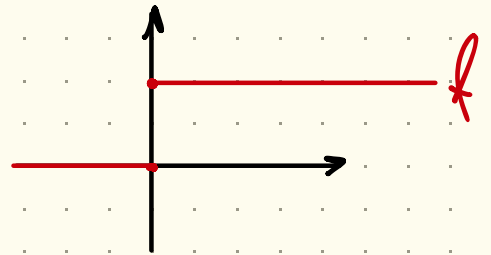
SE $f: I \rightarrow \mathbb{R}$ É CONTINUA, ALLORA

f AMMETTE UNA PRIMITIVA F

• OSSERVAZIONE. DERIVABILE $\Rightarrow$ CONT. $\Rightarrow$ AMMETTE PRIMITIVA

Es. $f: \mathbb{R} \rightarrow \mathbb{R}$

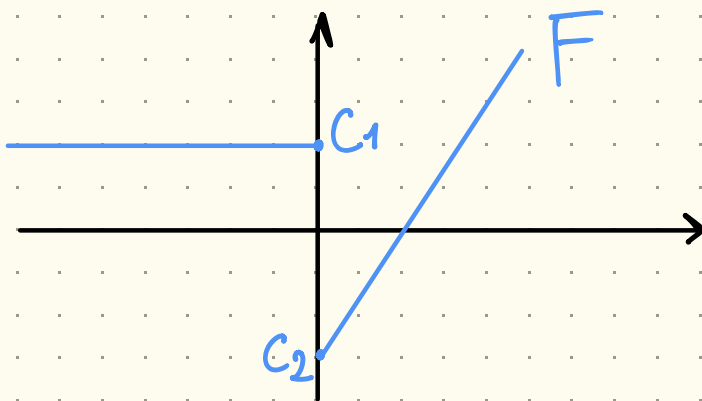
$$f(x) = \begin{cases} 0 & x < 0 \\ 1 & x > 0 \end{cases}$$



CERCO UNA PRIMITIVA DI f :

$$F(x) = \begin{cases} C_1 & x < 0 \\ x + C_2 & x > 0 \end{cases}$$

F DERIVABILE $\Rightarrow F$ CONTINUA



$$C_1 = C_2 = C$$

$$F(x) = \begin{cases} C & x < 0 \\ x + C & x > 0 \end{cases}$$

F É UNA PRIMITIVA DI f ? NO, POCHÉ

F NON É DERIVABILE IN 0 !!

OSSERVAZIONE. f : DISCONTINUITA A SALTO

$$b) \int |x| dx = \begin{cases} 1 & x > 0 \\ -1 & x < 0 \end{cases}$$

$$f(x) = |x| = \text{CONTINUA}$$

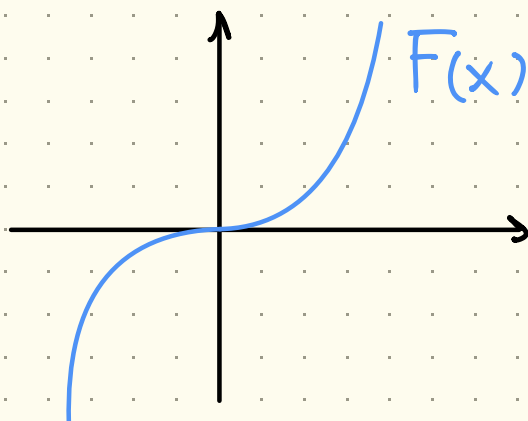
$\Rightarrow \exists$ PRIMITIVA F di f

DETERMINIAMOLA:

$$f(x) = \begin{cases} -x & x < 0 \\ x & x > 0 \end{cases}$$

$$F(x) = \begin{cases} -\frac{x^2}{2} & x < 0 \\ \frac{x^2}{2} & x > 0 \end{cases}$$

$$\left(\frac{x^2}{2}\right)' = \frac{1}{2} 2x = x$$



$$\Rightarrow \int |x| dx = F(x) + C$$

TEOREMA. (LINEARITA')

$f, g: I \rightarrow \mathbb{R}$ CONTINUA, $\alpha, \beta \in \mathbb{R}$, ALLORA

$$\int \alpha f(x) + \beta g(x) dx = \alpha \int f(x) dx + \beta \int g(x) dx$$

es) $\int \overset{\alpha}{3} \overset{f}{x} - \overset{\beta}{2} \frac{1}{x^2+1} dx$

$$= 3 \int x dx - 2 \int \frac{1}{x^2+1} dx$$

$$= 3 \frac{x^2}{2} - 2 \arctan x + C$$

SEGUE DA:

$$(\alpha F + \beta G)' = \alpha F' + \beta G'$$

$$\sin 2x = 2 \sin x \cos x$$

es) $\int (\sin x \cos x + \sin x) dx$

$$= \overset{A}{\int \sin x \cos x dx} + \overset{B}{\int \sin x dx}$$

$$A = \int \frac{\sin(2x)}{2} dx = \frac{1}{2} \int \sin(2x) dx$$

$$= \frac{1}{2} \left(\frac{-\cos(2x)}{2} \right)' = \frac{-\cos(2x)}{4} \cdot \left(\frac{-\cos(2x)}{2} \right)' =$$

$$= \frac{1}{2} (\cos(2x))' = \frac{1}{2} (-\sin 2x) \cdot 2$$

$$B = \int \sin x \, dx = -\cos x$$

PURTROPPO:

$$\int f(x)g(x) \, dx \neq \int f(x) \, dx \int g(x) \, dx$$

$$(FG)' = FG' + GF'$$

TEOREMA. (INTEGRAZIONI PER PARTI)

$f, g: I \rightarrow \mathbb{R}$ DERIVABILI CON f', g' CONTINUE

ALLORA

$$\int f(x)g'(x) \, dx = f(x)g(x) - \int f'(x)g(x) \, dx$$

$$f(x)g(x) = \int f(x)g'(x) + f'(x)g(x) \, dx$$

